

STACK OVERFLOW DEVELOPER SURVEY



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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- Data contextualization and analysis goal.
- Methodology description.
 - Data gathering.
 - Data analysis.
 - Data visualizations.
- Results presentation supported with graphs and trends.
- Discussion of overall findings and implications regarding the results previously exposed.
- Final conclusions of the carried-out research.

INTRODUCTION



- Stack Overflow's annual Developer Survey is the largest and most comprehensive survey of people who code around the world.
- Results don't represent everyone in the developer community evenly.
- Nearly 20,00 developers.
- Trends to predict where the developers are going.
- Characterization of developers around the globe.

METHODOLOGY

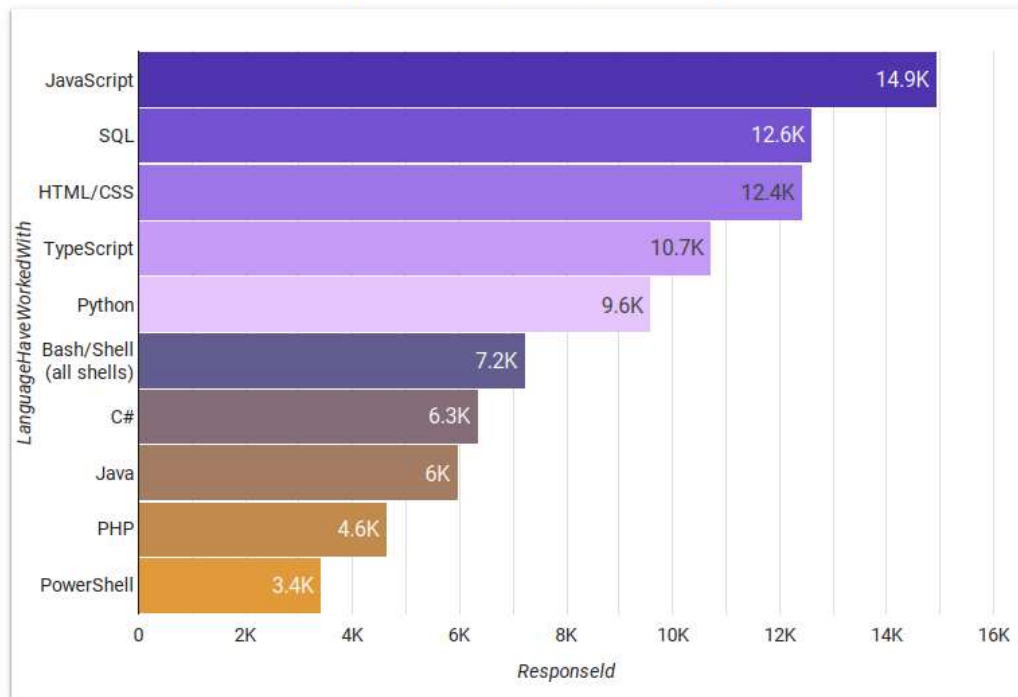


- Collect survey data & explore its content
 - Web Scraping.
 - APIs.
 - Request library.
- Data Wrangling
- Exploratory data analysis
 - Analyzing data distribution.
 - Handling outliers.
 - Correlations.
- Data Visualization
 - Highlight distribution of data, relationships, the composition and comparison of data.
- Dashboards

PROGRAMMING LANGUAGE TRENDS

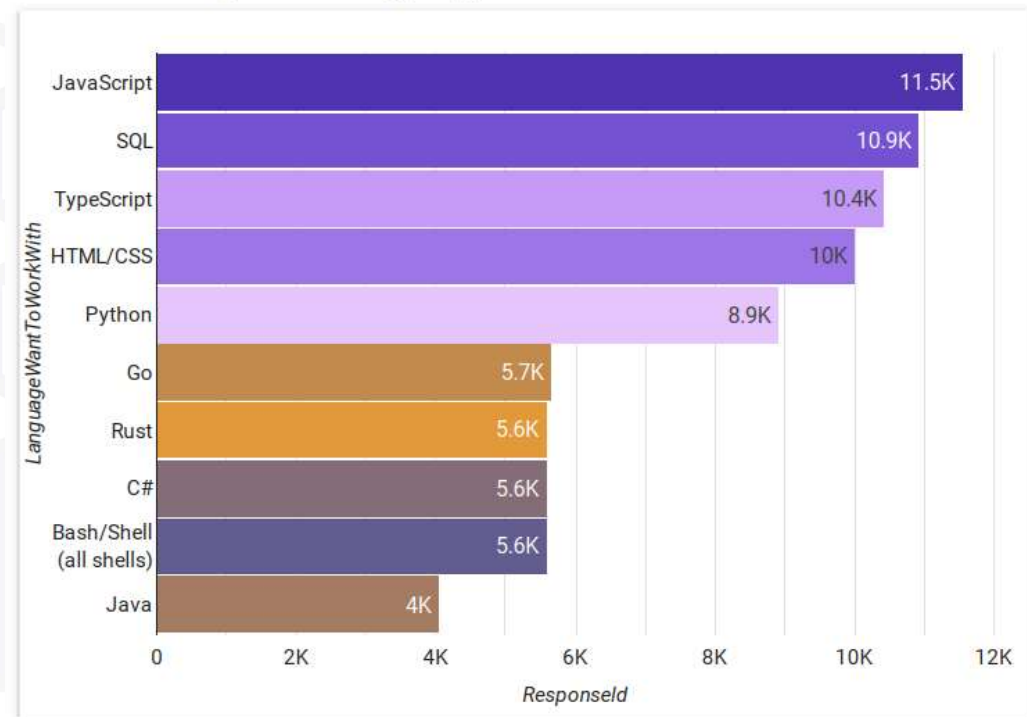
Current Year

Top 10 Languages Used



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript and SQL emerge as the most used programming languages among all respondents.
- TypeScript shows a significant increase in desirability.
- Python also maintains a significant presence.

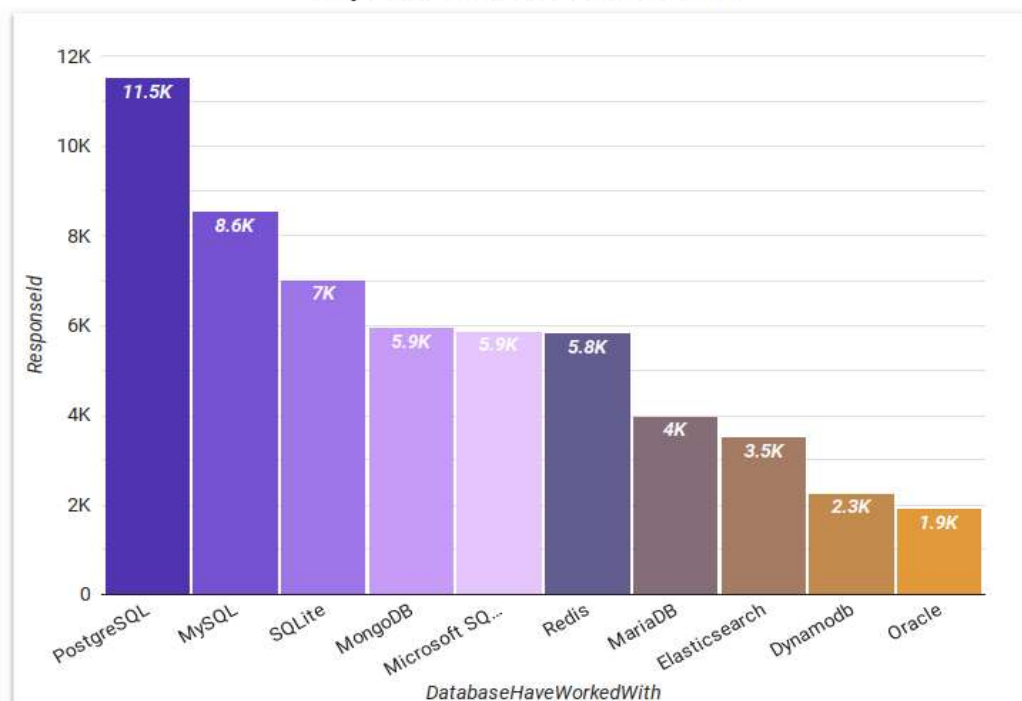
Implications

- JavaScript, SQL, and Python remain crucial. Mastering these languages is essential for developers.
- TypeScript's growing desirability suggests a bright future. Developers should consider learning it to enhance their skills.
- Increased interest in Go and Rust signals a shift towards performance and safety, particularly in cloud computing and systems programming

DATABASE TRENDS

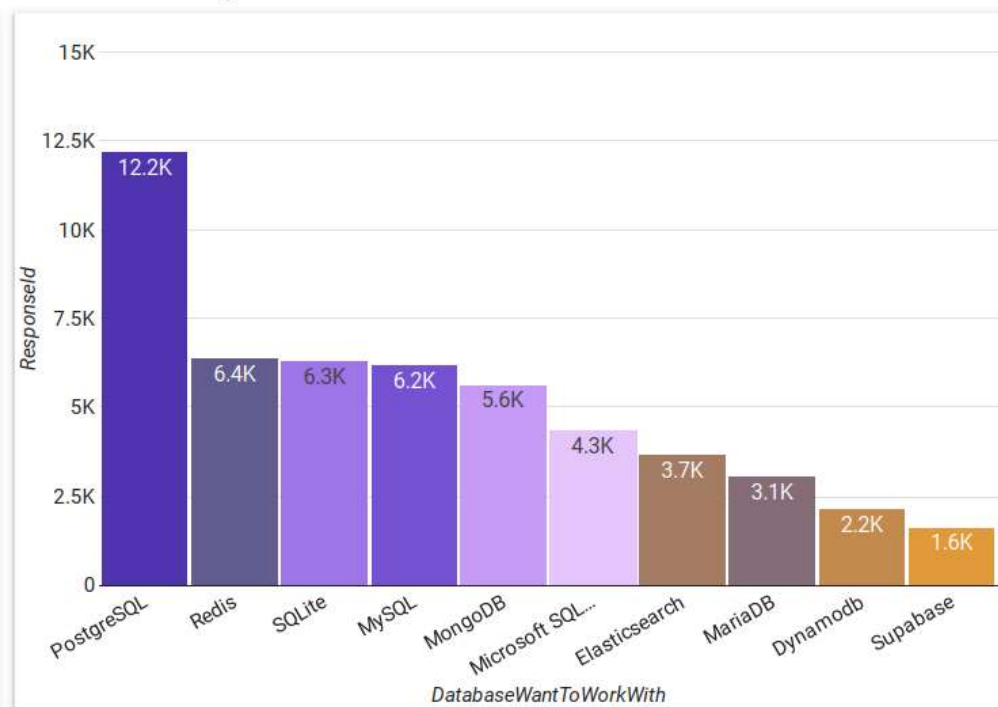
Current Year

Top 10 Database Used



Next Year

Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most used database
- Redis sees a significant increase in interest, becoming one of the top choices.
- SQLite, MySQL, and MongoDB maintain strong desirability among developers.

Implications

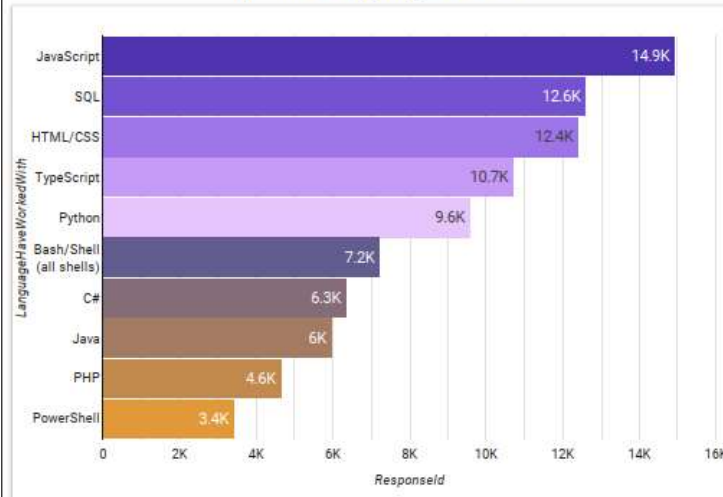
- PostgreSQL's consistent top ranking suggests its reliability, performance, and popularity among developers and organizations.
- The increased interest in Redis reflects its growing adoption for caching, real-time applications, and other high-performance needs.
- MySQL's continued popularity ensures that skills in this database will remain in demand.

DASHBOARD

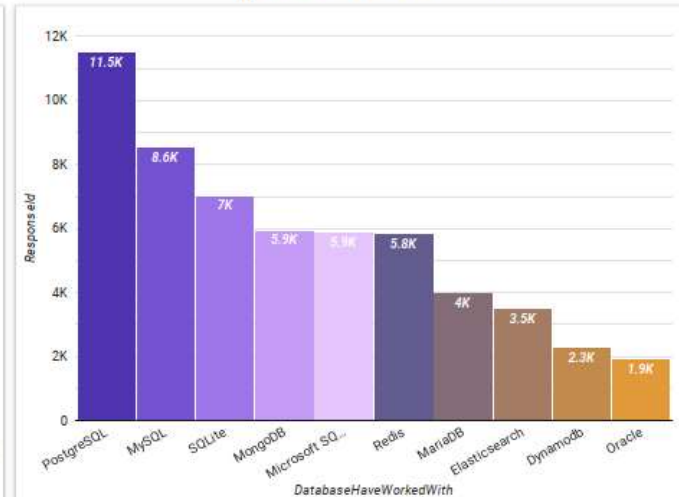


DASHBOARD TAB 1

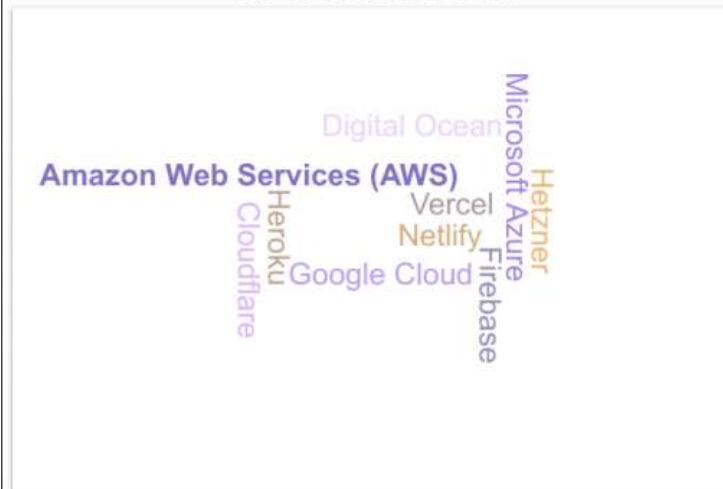
Top 10 Languages Used



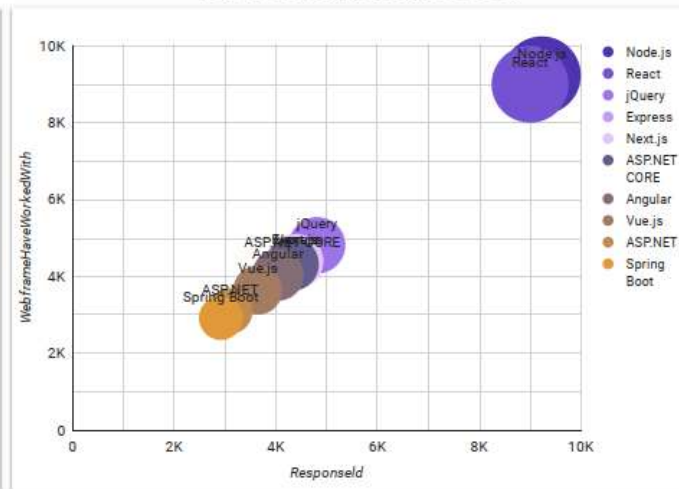
Top 10 Database Used



Top 10 Platforms Used

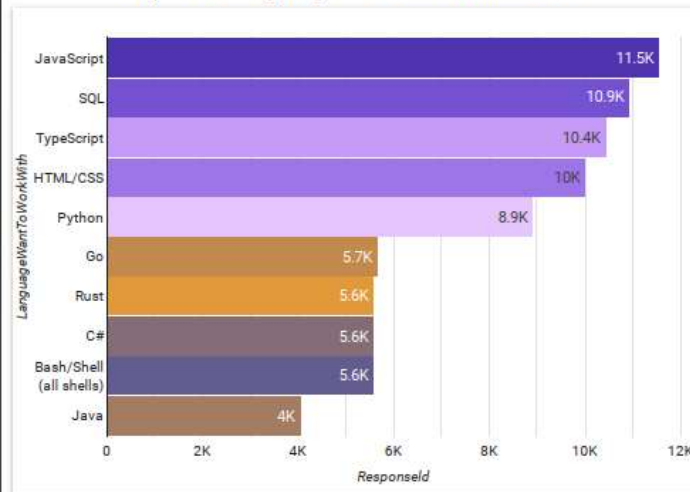


Top 10 Frameworks Used

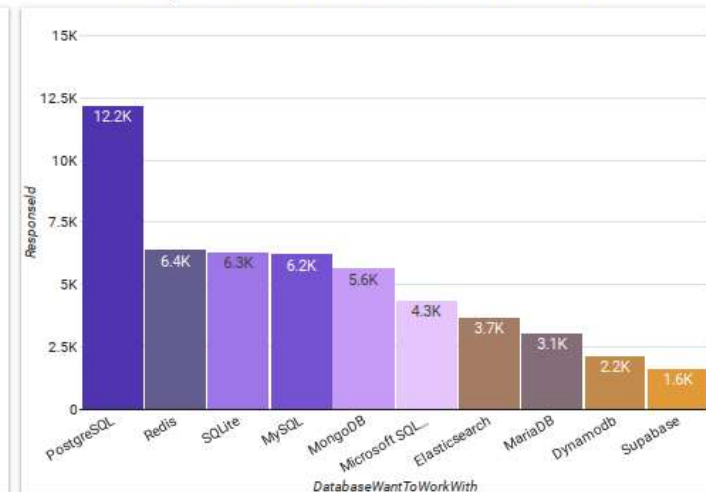


DASHBOARD TAB 2

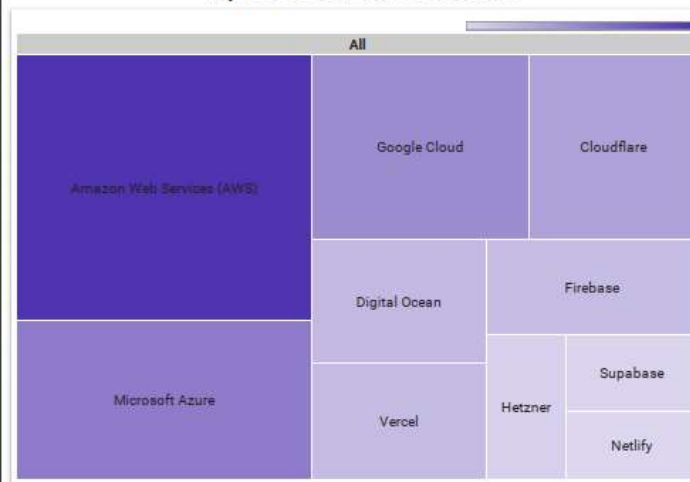
Top 10 Languages Desired Next Year



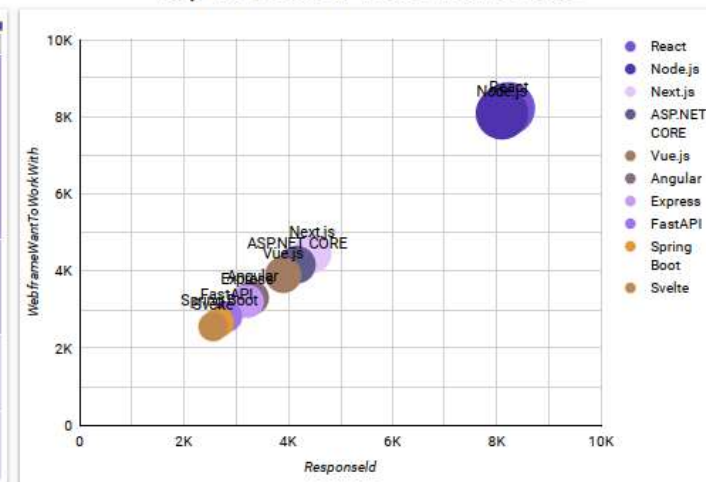
Top 10 Databases Desired Next Year



Top 10 Desired Platforms

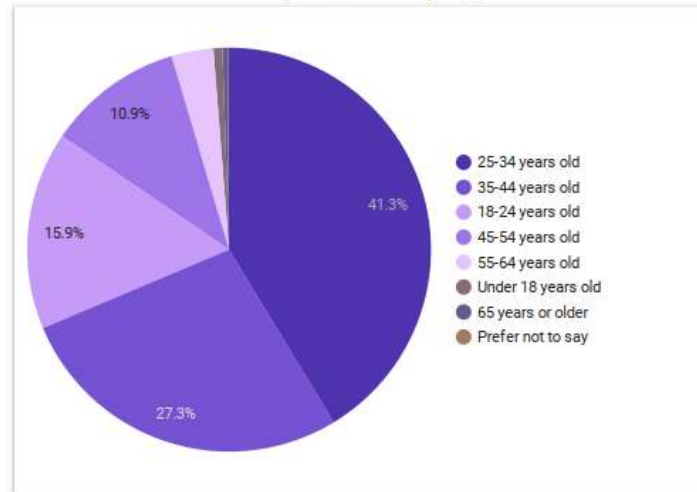


Top 10 Desired Web Frameworks

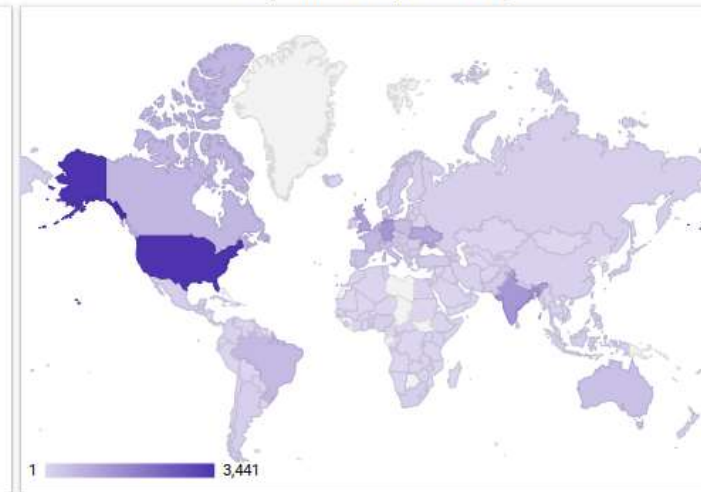


DASHBOARD TAB 3

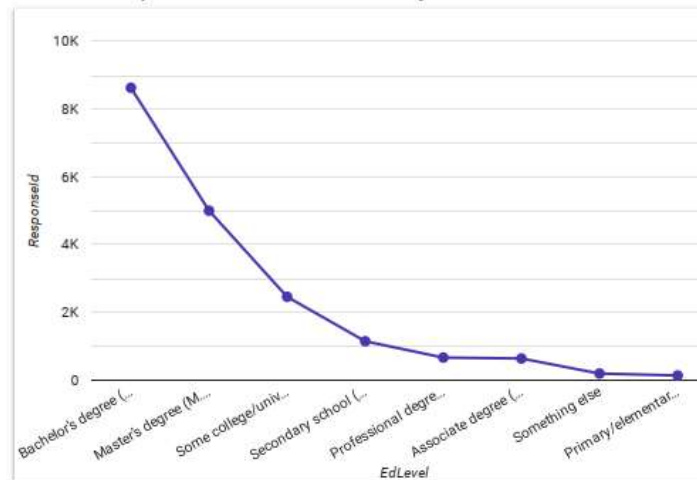
Respondents by Age



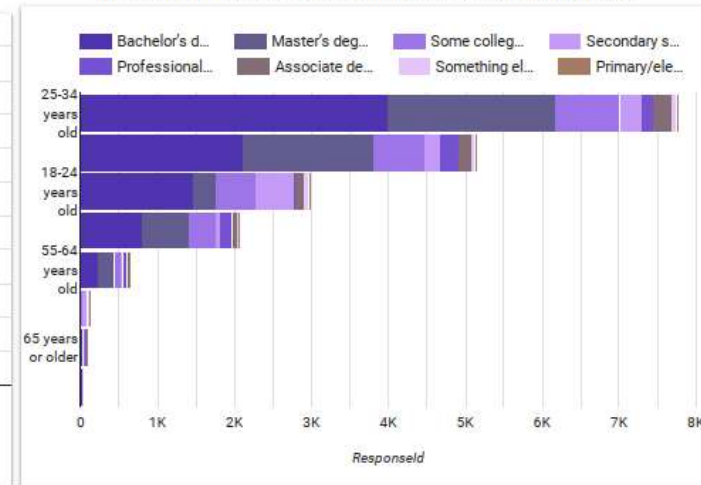
Responders by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, and Python remain top programming languages currently used and desired for the future, signaling their continued importance. Emerging languages like Rust and Go are gaining interest.
- PostgreSQL is the most used and desired database, with rising interest in Redis and MongoDB.
- Educational Background: A large proportion of respondents hold a bachelor's or master's degree, reflecting a well-educated professional base. Younger respondents (18-24 years) are predominantly early in their education or careers, while older respondents show varied educational levels, indicating diverse career trajectories.

Implications

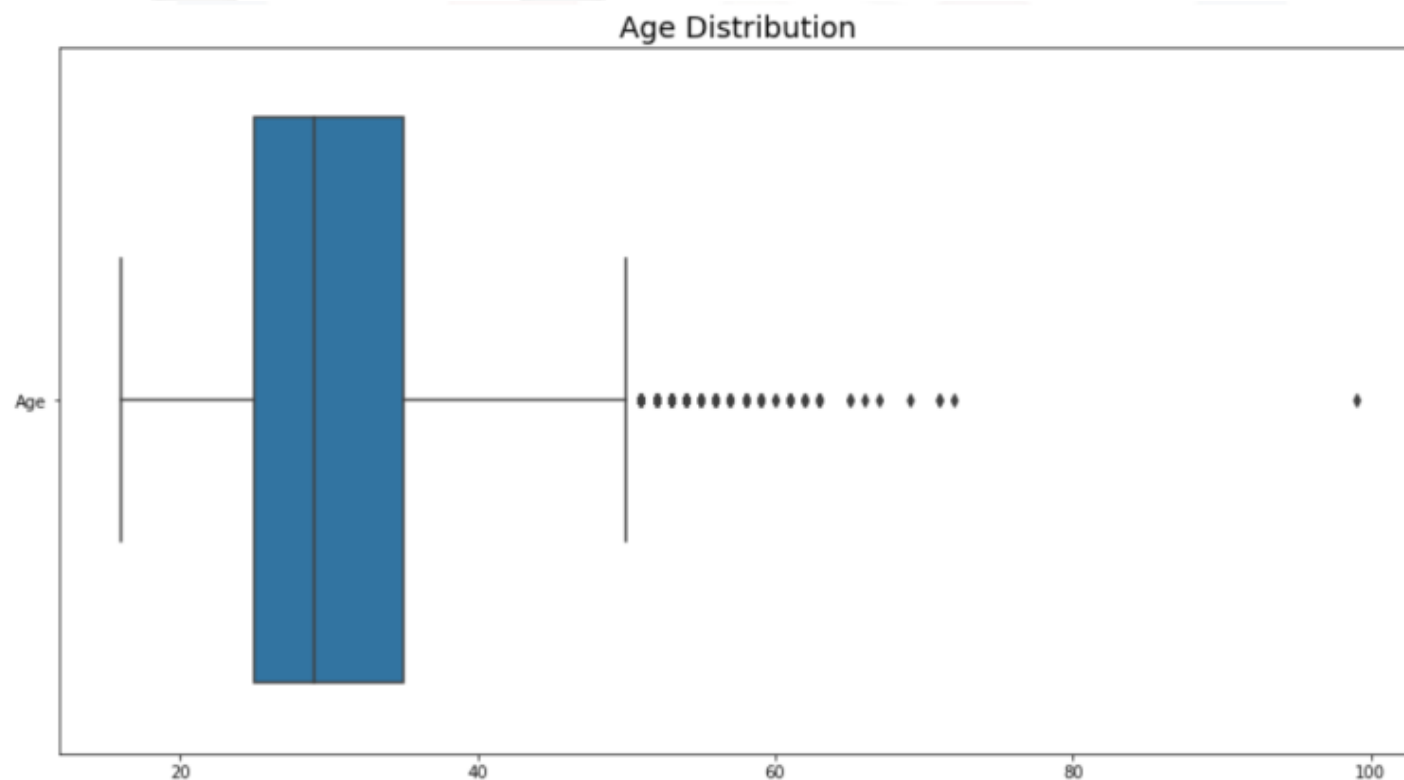
- Technological Preferences: The sustained demand for JavaScript, SQL, and Python indicates their reliability across projects, while emerging tools like Rust and Go signal evolving needs in performance and scalability. Organizations should invest in these newer technologies to stay competitive.
- The preference for PostgreSQL and cloud platforms reflects a growing emphasis on scalability and cloud-native development.
- Education and Global Outreach: The prevalence of higher education highlights the industry's reliance on skilled professionals, suggesting a need to expand training programs and certifications. Additionally, addressing low participation from underrepresented regions could foster diversity and innovation on a global scale.

CONCLUSION

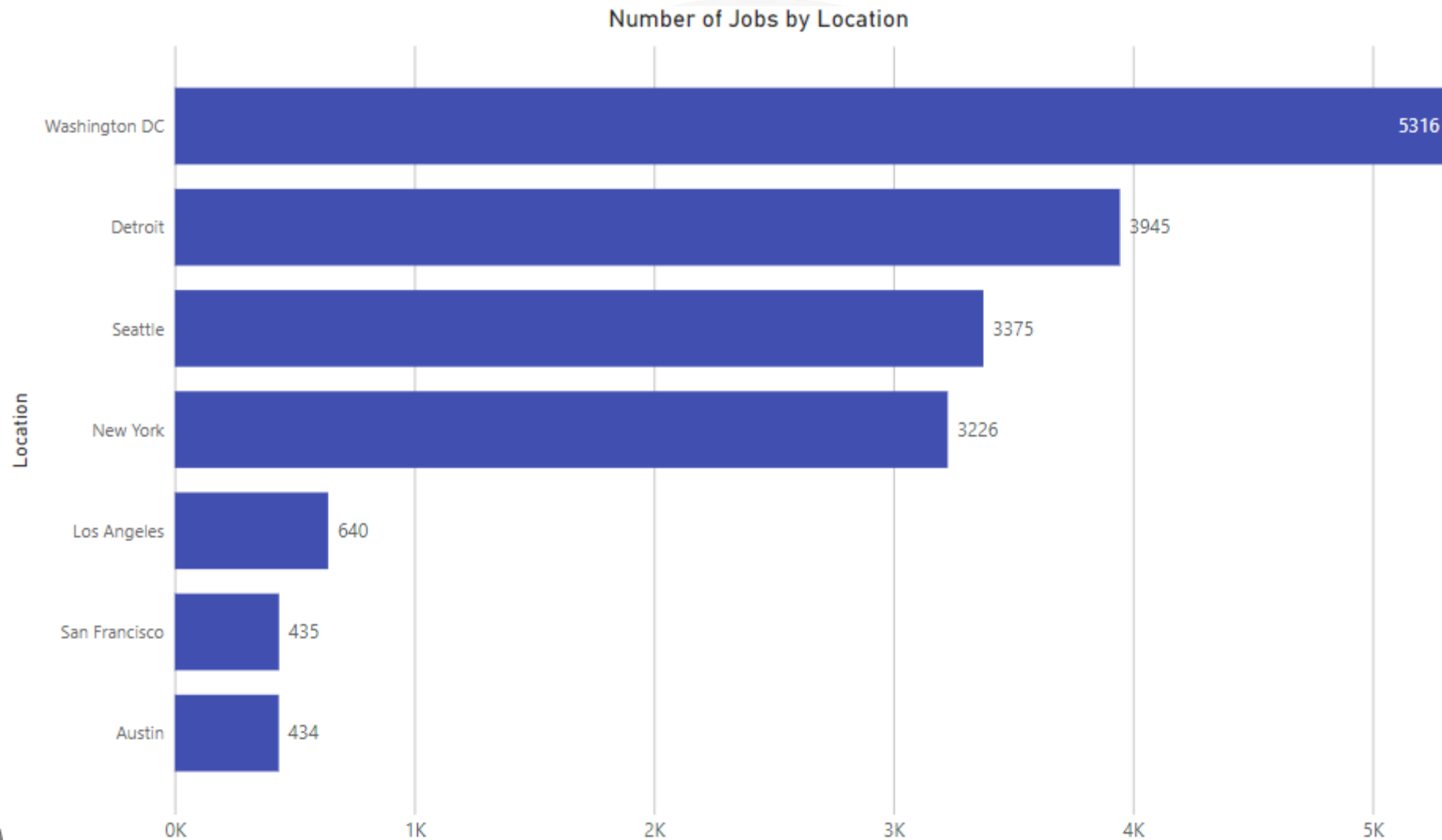


- Developers are people with very marked characteristics.
- A good idea of popularity trends of different tools, platforms and languages can be obtained.
- There is a job to be done to spread accessibility of this labor market to countries in development.

APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

